

Gabrielle Johnson's Interim CoDesign Canvas

The flow: patient is greeted by front desk staff and given a smart watch → watch captures vitals, location and a QR pairing code → nurse scans, accepts or rejects the vitals → model suggests an ESI level with SHAP reasoning → nurse accepts or declines with a logged reason → ESI acuity ordered queue → patient can scan their watch at the kiosk at any time for a status update → watch trend monitors during the wait, alerting nurses of deterioration.

Problem

Mercer General is a regional referral hospital for a Caribbean catchment of about 80,000 residents plus seasonal visitors. Its ED sees 70–95 patients on a normal day and 110–170 on a heavy one. Every patient is triaged by a nurse into ESI 1–5 in a 3–5 minute assessment but its paper based triage workflow contributes to estimated delays of 85% and 20% category disagreement.

Users

	Group	Characteristics	Needs
Secondary	Patients	Unwell, anxious, in pain; residents and seasonal visitors; wide range of age, language and digital literacy;	Assessed quickly and correctly; to know they have not been forgotten; to be re checked if they get worse
Primary	Triage nurses	Mobile; interrupted constantly; target ratios of 1:3 preferably but are not consistently achieved on heavy days	Accurate front door information; not to be flooded with false alarms; to keep authority over the triage decision

Other stakeholders:

1. Charge nurse — is the escalation target when an alert goes unacknowledged or a patient is unseen.
2. Registration clerks and a greeter — first nonclinical contact, and the people who route patients who cannot use the kiosk/watch.
3. ED physicians need the triage record and need to trust it.

Goals

	Shortterm	Longterm
Patients	Be seen in the right order; know what is happening during the wait	Not come to harm while waiting
Nurses	Complete vitals and assign esi level; know who is waiting, unseen and who has changed	Reducing esi level disagreements to less than 5% and improving retriaging efficiency, reducing interruptions from patients asking "how much longer?"

What the system does

Shortterm	Longterm
Capture vitals at checkin; suggest an ESI level with reasoning; hold the acuityordered queue; automate the waiting room re triaging	Reduce esi level disagreements; have a more efficient and safe ED; build a structured, queryable, deidentified dataset

What it does not do: assign the final ESI level, replace nurse triage, or replace the nurse's own visual sweep of the waiting room. ESI 1 is out of scope as it is clinically apparent at presentation and taken straight to resuscitation.

Why this rather than a screen or a person alone

- **Sensing over time** — continuous vitals during the wait rather than one reading can help prevent patient deterioration and can alert nurses in time
- **Mobility** — alerts reach a mobile nurse instead of assuming she is at a desk, and locate the patient among the waiting room and beyond.
- **Structured capture** — every encounter produces queryable data, which can be stored and help with Mercer's data collection process
- **Self service information** — patients can check their own status at the kiosk instead of approaching staff
- **Personalisation** — trend monitoring compares each patient to their own baseline rather than a fixed threshold.

- **Deliberately not a social agent** — no conversational or embodied persona. A human keeps the ESI decision and patient contact.

Ethics

Area	View	Problem	Solution
Physical safety	User	The watch sits on a limb that may have a wound, dressing etc	Contraindication check at issue; staff will apply and remove when necessary.
	System	The device could report a blood pressure value that it cannot measure reliably.	The watch is for passive sensing only; nurses must use their own judgement and can fall back on traditional methods to capture vitals if they desire
Clinical validity for this population	User	Mercer has no separate paediatric or obstetric ED, hence children and pregnant patients are triaged in the same space	Age captured at check-in; paediatric and obstetric patients are routed straight to a nurse
	System	The model was trained on adult data but would emit a confident number for a child regardless as it has no way to know it is out of scope	Collect age and ensure that no level is generated for patients under 18 or for obstetric patients.
Data security	User	Continuous vitals and indoor location are captured; Patients	At issue, patients are told plainly what is collected and why.

Area	View	Problem	Solution
		have little practical ability to refuse monitoring; a QR code on a watch can be photographed by a bystander	Location is captured only while the watch is issued, only within the department, deleted after the encounter and never used to monitor behaviour. QR tokens are short-lived and rotate frequently
	System	The system gathers sensitive patient data	Data governance requirements are required before deployment.
Sensory accessibility	User	A visual-only signal excludes blind and low vision patients; deaf patients cannot hear their name called; and many ED patients are temporarily impaired, or in too much pain to read	Two or three vibration patterns demonstrated at issue; every state carries an icon, a word and a motion pattern, not colour alone
	System	An ambient display is useless to a patient who cannot see it	All deterioration alerts, including the patient's location, are sent directly to the nurse's handheld device.
Information given to waiting patients	User	A quoted waiting time will be wrong on heavy days and	Status expressed as position or a range like "patients who are

Area	View	Problem	Solution
		during boarding; a missed time estimate erodes trust and is associated with frustration directed at staff, and a revised ESI would change a patient's position without explanation	more unwell are seen first" etc — never a countdown/timer
	System	A kiosk screen in a shared room can be read by whoever is standing behind the patient	No ESI level, no clinical detail and no reason shown on the kiosk

Environment

Where	Mercer ED ~20 seats waiting room. Mixed age and mixed acuity population. Live location coverage must reach the fasttrack area, corridors etc
When	24/7, Load rises after long weekends and through tourist high season
Users	Triage nurses
Secondary users	Patients; charge nurse; registration clerks; ED physicians
Simultaneous users	Assignment must be weighted by acuity rather than headcount: two ESI 2 patients are more work than five ESI 4s; Each user has access only to their dashboard

Data collected	Vitals, indoor location, chief complaint, user_id and every accept/decline with reason.
Sensors and signals	BLE or UWB anchors for indoor position, ambient amber attention state on the watch; vibration on the handheld devices, distinguishable from routine traffic
Connected systems	There is no electronic patient record to integrate with. The system must therefore stand alone and function with no IT dependency.

Workflow constraints

1. Triage has 3–5 minutes and a narrow data slice.
2. A nurse pulled away mid assessment must be able to resume without reentry.

Devices and deployment

1. No existing handheld fleet to inherit, so hardware, charging rotation, disinfection between patients and wrist size range are all new standing costs.
2. A worn device contributing to triage decisions must be a regulated medical device.